

Infinite Impulse Response

Filters :-

IIR $\rightarrow$ infinite no. of coefficients. $h(n) = a^n u(n)$ FIR $\rightarrow$ finite no. of coefficients. $h(n) = a^n [u(n) - u(n-1)]$

IIR

- * It has feedback loop in the design.

- * Recursive filter.

- * O/P, $y(n) = x(n) + x(n-1) + y(n-1)$

O/P of filter depends upon int i/p, delayed i/p, past output.

If past output it has feedback.

FIR

- * There is no feedback loop.

- * Non-recursive filter is another name.

- * O/P, $y(n) = x(n) + x(n-1)$

It doesn't depend on past O/P, so no presence of feedback loop in the design.

Design of digital IIR filter: $[h(n)]$, $(or) h(z)$

- * Finding the expression of digital transfer function.

Techniques:

 $H(s)$ or $H_c(s)$

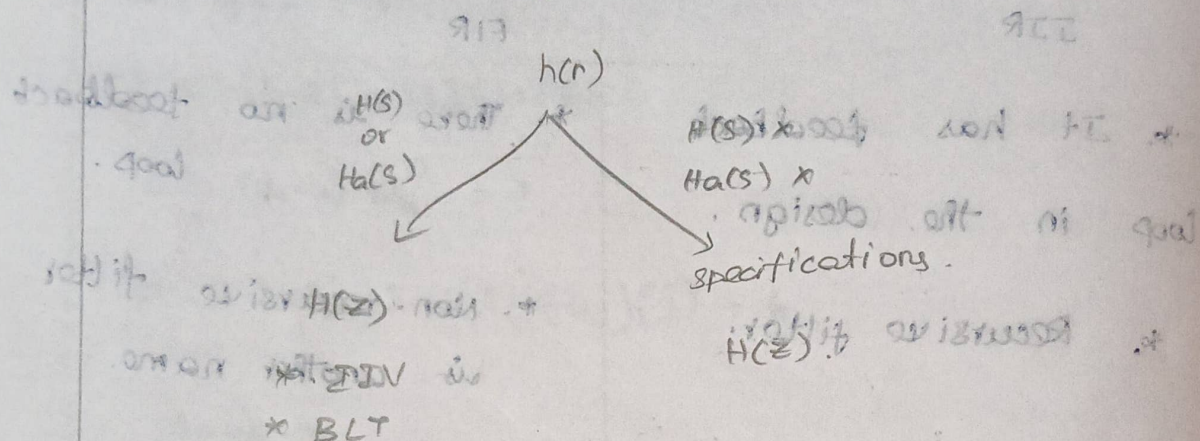
1. Analog transfer fn will be given, we need to find its equivalent digital

Transfer function, by using

* Impulse invariant method. (IIV)

* Bilinear Transform method. (BLT)

2). No analog T.F given instead of specification related to analog & digital T.F will be given.



Relation btw Laplace and z transform.

$$\rightarrow \text{LT}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \rightarrow \textcircled{1}$$

analog signal is sampled & apply laplace.

$$\text{LT}[x(nT)] = X(s) = \sum_{n=-\infty}^{\infty} x(nT) e^{-snT} \rightarrow \textcircled{2}$$

$$\rightarrow \text{ZT}[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \rightarrow \textcircled{3}$$

$$\text{eqn } \textcircled{2} \Rightarrow X(s) = \sum_{n=-\infty}^{\infty} x(nT) (e^{sT})^{-n} \rightarrow \textcircled{4}$$

now compare 3 & 4.

$$X(s) = X(z) \quad \text{at} \quad z = e^{sT}$$

$$s = \sigma + j\omega$$

$$z = re^{j\omega} \quad (\text{polar form})$$

where, $r \rightarrow$ magnitude of z

$$1T = zT, \quad \text{at} \quad z = e^{sT}$$

$$re^{j\omega} = e^{(\sigma + j\omega)T}$$

$$re^{j\omega} = e^{\sigma T} \cdot e^{j\omega T}$$

$$r = e^{\sigma T}$$

$$e^{j\omega} = e^{j\omega T}$$

digital frequency

analog frequency

$$\Rightarrow \omega = \Omega T$$

3 cases:

$$1. \sigma = 0$$

$$2. -\infty < \sigma < 0$$

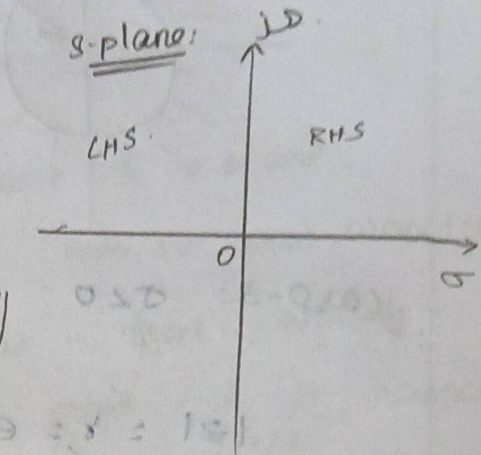
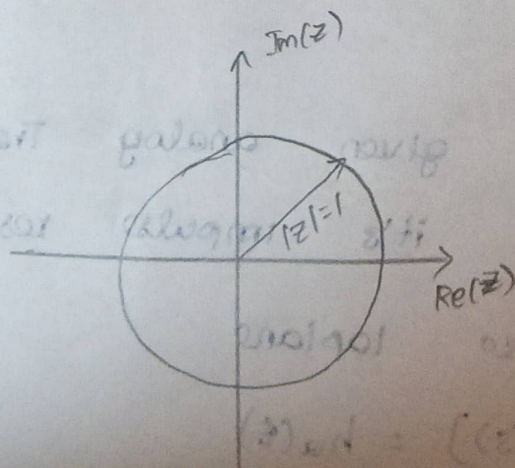
$$3. 0 < \sigma < \infty$$

$\rightarrow$ For stable analog filter all poles should be in left side of s -plane.

Case-1: $\sigma = 0$

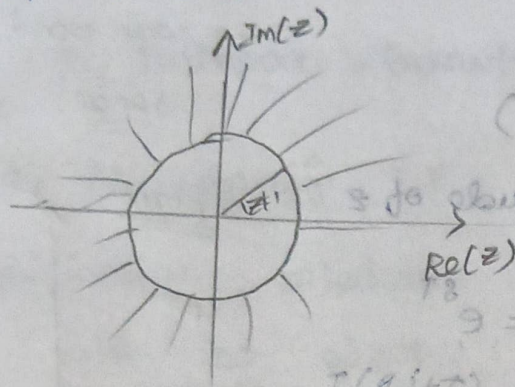
$$|z| = r = e^{\sigma T} = e^{0 \cdot T} = 1$$

$\sigma = 0 \Rightarrow z$ is mapped on the unit circle.



Case - 1: $\sigma > 0$ to $(s) \times = (z) \times$

$$|z| = r = e^{\sigma T} = \infty$$

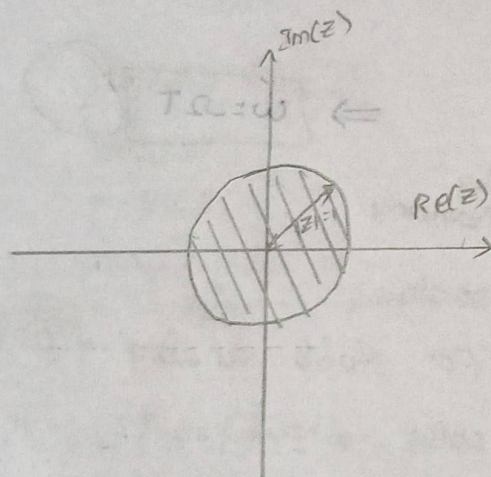


(entire RHS)

$\sigma > 0$ is mapped outside the unit circle.

Case - 2: $\sigma < 0$

$$|z| = r = e^{\sigma T} = e^{-\infty} = 0$$



$\sigma < 0$, entire LHS of s-plane

is mapped inside the unit circle.

Impulse invariance method.

$$H_a(s) \longrightarrow H(z)$$

$$H(s) = \frac{Y(s)}{X(s)}$$

Step-1: From the given analog Transfer function find its impulse response.

→ apply Inverse Laplace

$$\mathcal{L}^{-1} [H_a(s)] = h_a(t)$$

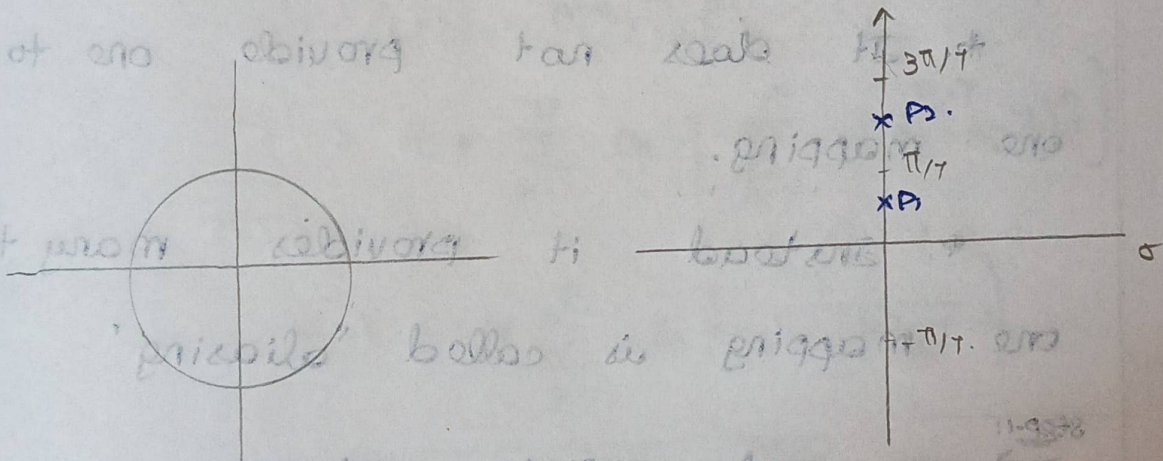
Step-1: Sampling. at $t=T$. $\Rightarrow$ to get the equivalent digital impulse for $h_a(t)$

$$h_a(t) \longrightarrow h_a(nT)$$

Step-2: Apply Z-transform.

$$h_a(nT) \longrightarrow H(z)$$

- * In mapping b/w s-plane & z-plane, it should provide confirmed (one to one) mapping.
- * It should map the two analog poles with one digital pole. (if it happens that is called aliasing)



The two poles,

$$s_1 = \sigma + j\omega_1$$

$$s_2 = \sigma + j\left(\omega_1 + \frac{2\pi}{T}\right)$$

real axis is same but Im axis different

Second analog pole is located on imaginary axis with distance of $\frac{2\pi}{T}$ from the 1st pole.

$$z_1 = e^{s_1 T} = e^{(\sigma + j\omega_1) T} = e^{\sigma T} \cdot e^{j\omega_1 T}$$

$$z_2 = e^{s_2 T} = e^{(j + j\omega_2) T} = e^{(\sigma + j[\omega_1 + \frac{2\pi}{T}]) T}$$

$$= e^{\sigma T} \cdot e^{j\omega_1 T} \cdot e^{j\frac{2\pi}{T} T}$$

$$= e^{\sigma T} \cdot e^{j\omega_1 T} \cdot 1$$

Disadvantage:

$$= z_1$$

Here two different analog poles are mapped to one digital pole.

* It does not provide one to one mapping.

* Instead it provides many to one mapping is called 'aliasing'

Step-1:

→ we need express in terms of summation of simple poles.

$$H_A(s) = \frac{C_1}{s - p_1} + \frac{C_2}{s - p_2} + \dots + \frac{C_N}{s - p_N}$$

→ Generalized expression given analog.

$$H_A(s) = \sum_{k=1}^N \frac{C_k}{s - p_k}$$

$$\mathcal{L}^{-1}[H_A(s)] = h_A(t) = \mathcal{L}^{-1}\left[\sum_{k=1}^N \frac{C_k}{s - p_k}\right] = \sum_{k=1}^N C_k e^{p_k t}$$

Step-2:

→ replace $t = nT$ (sampling process)

$$h_a(nT) = \sum_{k=1}^N \frac{C_k e^{P_k nT}}{1}$$

Step-3:

→ apply z-transform.

$$Z.T [h_a(nT)] = \sum_{n=0}^{\infty} h_a(nT) z^{-n}$$

$$H(z) = \sum_{n=0}^{\infty} \left[\sum_{k=1}^N C_k e^{P_k nT} z^{-n} \right]$$

$$= \sum_{n=0}^{\infty} \left[\sum_{k=1}^N (C_k) (e^{P_k T} z^{-1})^n \right]$$

$$= \sum_{k=1}^N C_k \left[\sum_{n=0}^{\infty} (e^{P_k T} z^{-1})^n \right]$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

$$= \sum_{k=1}^N C_k \left[\frac{1}{1 - e^{P_k T} z^{-1}} \right]$$

$$H(z) = \sum_{k=1}^N \frac{C_k}{1 - e^{P_k T} z^{-1}}$$

$$H_a(s) = \sum_{k=1}^N \left[\frac{C_k}{s - P_k} \right] \xrightarrow{\text{IV}} H(z) = \sum_{k=1}^N \frac{C_k}{1 - e^{P_k T} z^{-1}}$$

$$\frac{H_a(s)}{s - P_k} = \frac{1}{s - P_k}$$

$$\frac{H(z)}{1 - e^{P_k T} z^{-1}} = \frac{1}{1 - e^{P_k T} z^{-1}}$$

$$\frac{1}{s - P_k} \xrightarrow{\text{IV}} \frac{1}{1 - e^{P_k T} z^{-1}}$$

1. Convert the given analog T.F. $H_a(s) = \frac{2}{(s+1)(s+2)}$ into its equivalent digital IIR filter using impulse invariance method.

Sol:

given, $H_a(s) = \frac{2}{(s+1)(s+2)} \rightarrow \textcircled{1}$.

Step-1:

Apply partial fraction.

$$\frac{2}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}.$$

$$2 = A(s+2) + B(s+1)$$

Put $s = -2$ $\Rightarrow$ $2 = A(1)$

$$2 = B(-1)$$

$B = -2$ $\Rightarrow$ $A = 2$

$$H_a(s) = \frac{2}{(s+1)} - \frac{2}{(s+2)} \rightarrow \textcircled{2}$$

(Apply inverse Laplace.

$$\mathcal{L}^{-1}[H_a(s)] = h_a(t) = \mathcal{L}^{-1}\left[\frac{2}{s+1}\right] - \mathcal{L}^{-1}\left[\frac{2}{s+2}\right]$$

$$h_a(t) = 2e^{-t} - 2e^{-2t}$$

Step-2: Sampling $t = nT$.

$$h_a(nT) = 2(e^{-nT} - e^{-2nT})$$

Step-3:

z transform is applied.

$$Z.T [h_a(nT)] = \sum_{n=0}^{\infty} h_a(nT) z^{-n}$$

$$= \sum_{n=0}^{\infty} 2(e^{-nT} - e^{-2nT}) z^{-n}$$

$$= 2 \sum_{n=0}^{\infty} (e^{-nT} - e^{-2nT}) z^{-n}$$

$$= 2 \left[\frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-2T} z^{-1}} \right]$$

(or)

$$\frac{1}{s+1} \longrightarrow \frac{1}{1 - e^{-T} z^{-1}}$$

$$\frac{1}{s+2} \longrightarrow \frac{1}{1 - e^{-2T} z^{-1}}$$

$$H(z) = 2 \left[\frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-2T} z^{-1}} \right]$$

here, $T \rightarrow$ sampling period.

here T is not given in question so take $T=1$

assume, $T=1 \text{ sec.}$

$$H(z) = 2 \left[\frac{1}{1 - e^{-1} z^{-1}} - \frac{1}{1 - e^{-2} z^{-1}} \right]$$

$$= 2 \left[\frac{1}{1-0.367z^{-1}} - \frac{1}{1-0.135z^{-1}} \right]$$

$$= 2 \left[\frac{1-0.135z^{-1} - 1 + 0.367z^{-1}}{(1-0.367z^{-1})(1-0.135z^{-1})} \right]$$

$$= 2 \left[\frac{0.232z^{-1}}{1-0.135z^{-1}-0.367z^{-1}+0.049z^{-2}} \right]$$

$$H(z) = \frac{0.464z^{-1}}{1-0.502z^{-1}+0.049z^{-2}}$$

Realization:

* The filter has to be in implemented form.

to design filter we require,

* delay

* summer

* multiplier

2. Find $H(z)$ expression if $H_a(s) = \frac{s+a}{(s+a)^2 + b^2}$ using IIV method.

Sol:-

$$\mathcal{L}^{-1}[H_a(s)] = e^{-at} \cos(bt)$$

sampling.

$$h_a(nT) = e^{-anT} \cos(bnT)$$

apply z-transform:

$$H(z) = z^{-T} [h_a(nT)] = \sum_{n=0}^{\infty} h_a(nT) z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-anT} \cos(bnT) z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-anT} \left[\frac{e^{jbnT} + e^{-jbnT}}{2} \right] z^{-n}$$

$$= \frac{1}{2} \left[\sum_{n=0}^{\infty} z^{-n} e^{-anT} e^{jbnT} + \sum_{n=0}^{\infty} z^{-n} e^{-anT} e^{-jbnT} \right]$$

$$= \frac{1}{2} \left[\sum_{n=0}^{\infty} \left(z^{-1} e^{-aT} e^{j b T} \right)^n + \sum_{n=0}^{\infty} \left(z^{-1} e^{-aT} e^{-j b T} \right)^n \right]$$

$$= \frac{1}{2} \left[\frac{1}{1 - e^{-aT} e^{j b T} z^{-1}} + \frac{1}{1 - e^{-aT} e^{-j b T} z^{-1}} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1 - e^{-(a-jb)T} z^{-1}} + \frac{1}{1 - e^{-(a+jb)T} z^{-1}} \right]$$

$$= \frac{1}{2} \left[\frac{1 - e^{-(a+jb)T} z^{-1} + 1 - e^{-(a-jb)T} z^{-1}}{1 - e^{-(a+jb)T} z^{-1} - e^{-(a-jb)T} z^{-1} + e^{-(a-jb)T} z^{-1} e^{-(a+jb)T} z^{-1}} \right]$$

$$= \frac{1}{2} \left[\frac{2 - (e^{-aT} e^{-jbT} z^{-1} + e^{-aT} e^{jbT} z^{-1})}{1 - e^{-aT} e^{-jbT} z^{-1} - e^{-aT} e^{jbT} z^{-1} + z^{-2} e^{-2aT}} \right]$$

$$H(z) = \frac{1}{2} \left[\frac{2 - e^{-aT} [e^{-jbT} + e^{jbT}] z^{-1}}{1 - e^{-aT} [e^{-jbT} + e^{jbT}] z^{-1} + e^{-2aT} z^{-2}} \right]$$

$$= \frac{1}{2} \left[\frac{2 - 2e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}} \right]$$

$$H(z) = \frac{2}{2} \left[\frac{1 - e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}} \right]$$

$$H(z) = \frac{1 - e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}}$$

3. Find $H(z)$ if $H_a(s) = \frac{b}{(s+a)^2 + b^2}$

Soln

$$\mathcal{L}^{-1}[H_a(s)] = \mathcal{L}^{-1} \left[\frac{b}{(s+a)^2 + b^2} \right] = e^{-at} \sin(bt)$$

$$h_a(nT) = e^{-anT} \sin(bnT)$$

$$z[h_a(nT)] = z[e^{-anT} \sin(bnT)]$$

$$= \sum_{n=0}^{\infty} e^{-anT} \sin(bnT) z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-anT} \left[\frac{e^{-jb nT}}{2j} - \frac{e^{jb nT}}{-2j} \right] z^{-n}$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} e^{-ant} e^{jbnt} z^{-n} - \sum_{n=0}^{\infty} e^{-ant} e^{-jbnt} z^{-n} \right]$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} (e^{-(a-jb)T} z^{-1})^n - \sum_{n=0}^{\infty} (e^{-(a+jb)T} z^{-1})^n \right]$$

$$= \frac{1}{2j} \left[\frac{1}{1 - e^{-(a-jb)T} z^{-1}} - \frac{1}{1 - e^{-(a+jb)T} z^{-1}} \right]$$

$$= \frac{1}{2j} \left[\frac{1 - e^{-(a+jb)T} z^{-1}}{1 - e^{-(a-jb)T} z^{-1} - e^{-(a+jb)T} z^{-1} + e^{-(a-jb)T} z^{-2}} \right]$$

$$= \frac{1}{2j} \left[\frac{-e^{-aT} e^{-jbT} z^{-1} + e^{-aT} e^{jbT} z^{-1}}{1 - e^{-aT} z^{-1} (e^{-jbT} + e^{jbT}) + e^{-2aT} z^{-2}} \right]$$

$$= \frac{1}{2j} \left[\frac{e^{-aT} z^{-1} (e^{jbT} - e^{-jbT})}{1 - e^{-aT} z^{-1} (e^{-jbT} + e^{jbT}) + e^{-2aT} z^{-2}} \right]$$

$$= \frac{1}{2j} \left[\frac{2j e^{-aT} z^{-1} \sin bT}{1 - 2 \cos(bT) e^{-aT} z^{-1} + e^{-2aT} z^{-2}} \right]$$

$$H(z) = \frac{e^{-aT} z^{-1} \sin bT}{1 - 2 \cos(bT) e^{-aT} z^{-1} + e^{-2aT} z^{-2}}$$

$$\parallel$$

4. Design 3rd order Butterworth digital filter using IIV technique. Assume $T=1800$.

Transfer function of 3rd order Butterworth filter.

Sol:

$$H_a(s) = \frac{1}{(s+1)(s^2+s+1)}$$

Apply partial fraction.

$$H_a(s) = \frac{A}{s+1} + \frac{Bs+C}{s^2+s+1}$$

$$\frac{1}{(s+1)(s^2+s+1)} = \frac{A(s^2+s+1) + (Bs+C)(s+1)}{(s+1)(s^2+s+1)}$$

$$1 = A(s^2+s+1) + (Bs+C)(s+1)$$

put $s=-1$ put $s=0$

$$1 = A(1-1+1) \Rightarrow A=1$$

$$1 = A+C$$

$$1 = 1+C \Rightarrow C=0$$

equating s^2 coefficients.

$$0 = A+B$$

$$0 = 1+B \Rightarrow B=-1$$

$$B=-1$$

$$H_a(s) = \frac{1}{s+1} - \frac{s}{s^2+s+1}$$

$$H_a(s) = \frac{1}{s+1} - \frac{\left(\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right) - \frac{\frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$z[h_a(t)] = e^{-t} - \left(e^{-\frac{1}{2}t} \cos \frac{\sqrt{3}}{2} t - \frac{1}{\sqrt{3}} e^{-\frac{1}{2}t} \sin \frac{\sqrt{3}}{2} t \right)$$

$$h(nT) = e^{-nT} - \left(e^{-\frac{1}{2}nT} \cos \frac{\sqrt{3}}{2} nT - \frac{1}{\sqrt{3}} e^{-\frac{1}{2}nT} \sin \frac{\sqrt{3}}{2} nT \right)$$

$$\therefore T = 1 \text{ sec.}$$

$$\left[e^{-anT} \cos(bnT) \xrightarrow{z^{-1}} \frac{1 - e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}} \right.$$

$$\left. e^{-bnT} \sin(bnT) \xrightarrow{z^{-1}} \frac{e^{-aT} \sin(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}} \right]$$

$$\xrightarrow{s+1} \frac{1}{1 - e^{-1} z^{-1}}$$

$$H(z) = \frac{1}{(-0.367 z^{-1})} \left[\frac{1 - (0.606)(0.647) z^{-1}}{1 - 2(0.606)(0.647) z^{-1} + 0.367 z^{-2}} \right.$$

$$\left. - \frac{1}{\sqrt{3}} \frac{(0.606)(0.761) z^{-1}}{1 - 2(0.606)(0.647) z^{-1} + 0.367 z^{-2}} \right]$$

$$= \frac{1}{1 - 0.367 z^{-1} - 1} \frac{1 - 0.393 z^{-1} - 0.066 z^{-1}}{1 - 0.784 z^{-1} + 0.367 z^{-2}}$$

$$H(z) = \frac{1}{1 - 0.367 z^{-1}} - \frac{1 - 0.658 z^{-1}}{1 - 0.784 z^{-1} + 0.367 z^{-2}}$$

$$\cos\left(\frac{\sqrt{3}}{2}\right) = 0.647 \quad (\text{in radians})$$

$$\sin\left(\frac{\sqrt{3}}{2}\right) = 0.761$$

$$e^{-1/2} = 0.367$$

$$e^{-1} = 0.606$$

Relation b/w analog & digital frequency.

$$\omega = \Omega T$$

where, $\omega \rightarrow$ Digital frequency.

$\Omega \rightarrow$ analog frequency.

$T \rightarrow$ separation period.

BILINEAR TRANSFORMATION (BLT)

(unique)

mapping /

$\rightarrow$ Here one to one

relationship b/w analog to digital poles.

$\rightarrow$ Also known as conformal mapping.

$\rightarrow$ It does not provide aliasing.

Relations b/w s-plane and z-plane.

$$s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$$

$$j\Omega = \frac{2}{T} \left[\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right]$$

$$j\Omega = \frac{2}{T} \left[\frac{e^{j\omega/2} e^{-j\omega/2} (-e^{-j\omega/2} e^{-j\omega/2})}{e^{j\omega/2} e^{-j\omega/2} + e^{-j\omega/2} e^{-j\omega/2}} \right]$$

$$j\Omega = \frac{2}{T} \frac{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2} (e^{j\omega/2} + e^{-j\omega/2})}$$

$$\omega = \frac{\Omega}{T} \sin\left(\frac{\omega}{2}\right)$$

$$\Omega = \frac{\omega}{\sin\left(\frac{\omega}{2}\right)}$$

$$\Omega = \frac{\omega}{T} \tan\left(\frac{\omega}{2}\right)$$

$$\tan\left(\frac{\omega}{2}\right) = \frac{\Omega T}{2}$$

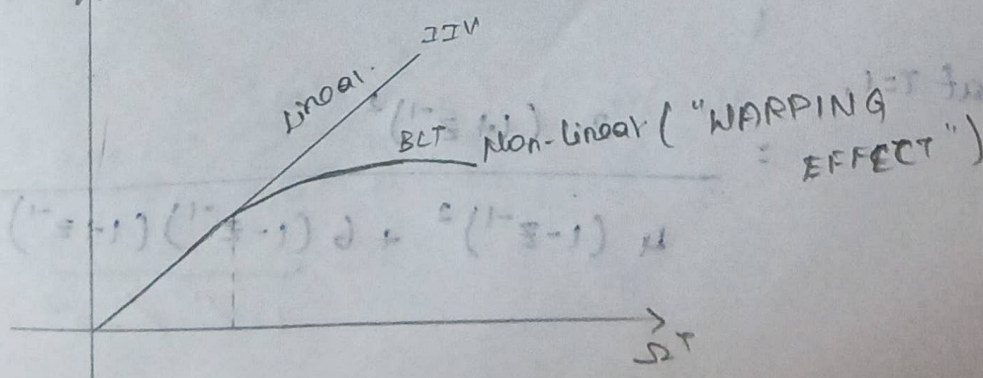
$$\omega = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right)$$

* In IIV, $\omega = \Omega T$ it shows linear relationship b/w digital & analog frequency.

* In BLT, up to some value of ΩT is linear after that it is not linear relationship.

* Due to non-linearity, we don't get proper frequency response, distortion occurs. "WARPING EFFECT"

$$\omega = \left(\frac{1 - \cos\left(\frac{\omega}{2}\right)}{1 + \cos\left(\frac{\omega}{2}\right)} \right) \frac{\Omega}{T} = \left(\frac{1 - \cos\left(\frac{\omega}{2}\right)}{1 + \cos\left(\frac{\omega}{2}\right)} \right) \frac{\Omega}{T}$$



* To avoid this effect, we need to find a pre-warping value so that we can maintain linear relationship.

→ design of DT is very easy.
Pre-warping frequency, $\frac{T\omega_c}{c} = \left(\frac{\omega}{c}\right)$ not

$$\omega_p = \frac{T\omega_c}{c} \tan\left(\frac{\omega_p}{2}\right)$$

① For the analog transfer function, $H(s) = \frac{2}{s^2 + 3s + 2}$, determine $H(z)$ using BLT.

Assume that $T = 1 \text{ sec}$, $T = 0.1 \text{ sec}$: pre-warp

sol: In BLT, that effect occur in

$$H(z) = H(s) \text{ at } s = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H(s) = \frac{2}{s^2 + 3s + 2}$$

$$H(z) = \frac{2}{\left\{ \left[\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + \frac{6}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2 \right\}}$$

at $T=1$,

$$= \frac{2(1+z^{-1})^2}{4(1-z^{-1})^2 + 6(1-z^{-1})(1+z^{-1}) + 2(1+z^{-1})^2}$$

$$1+z^{-1}-z^{-1}-z^{-2}$$

$$= \frac{2(1+z^{-2}+2z^{-1})}{4+4z^{-2}+8z^{-1}+6-6z^{-2}+2+2z^{-2}+4z^{-1}}$$

$$= \frac{2+2z^{-2}+4z^{-1}}{12+4z^{-1}} \quad \frac{1+z^{-2}+2z^{-1}}{2(3+z^{-1})}$$

$$H(z) = \frac{1+2z^{-1}+z^{-2}}{6-2z^{-1}}$$

Q. convert the analog filter to the system function $H_a(s) = \frac{1}{s^2+2s+1}$ into a digital filter using impulse invariance method.

Sol:-

$$H_a(s) = \frac{1}{s^2+2s+1} = \frac{1}{(s+1)^2} = \frac{1}{(s+1)(s+1)}$$

$$h_a(t) = e^{-t} t u(t)$$

$$h_a(nT) = e^{-nT} (nT) u(nT)$$

$$\therefore T=1 \text{ sec}$$

$$h_a(n) = n e^{-n} u(n)$$

applying Z transform

$$H(z) = \sum_{n=0}^{\infty} h_a(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} n e^{-n} u(n) z^{-n}$$

$$H(z) = \sum_{n=0}^{\infty} n e^{-n} z^{-n}$$

$$= \sum_{n=0}^{\infty} n (e^{-1} z^{-1})^n$$

$$= \frac{e^{-1} z^{-1}}{(1 - e^{-1} z^{-1})^2}$$

$$= \frac{0.3678 z^{-1}}{(1 - 0.3678 z^{-1})^2}$$

$$(1 - 0.3678 z^{-1})^2$$

$$H(z) = \frac{0.3678 z^{-1}}{1 - 0.7356 z^{-1} + 0.1352 z^{-2}}$$

(3)

$$H_a(s) = \frac{340.1}{(s + 0.1)^2 + 9}$$

$$h_a(t) = e^{-0.1t} \cos 3t$$

$$h_a(nT) = e^{-0.1nT} \cos 3nT$$

$$T = 1800$$

$$h_a(n) = e^{-0.1n} \cos 3n$$

$$H(z) = \frac{1 - e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}}$$

$$a = 0.1, b = 3, T = 1$$

$$H(z) = \frac{1 - e^{-0.1} \cos(3) z^{-1}}{1 - 2e^{-0.1} \cos(3) z^{-1} + e^{-2 \times 0.1} z^{-2}}$$

$$e^{-0.1} = 0.9048$$

$$\cos(3) = -0.9899$$

$$e^{-0.2} = 0.8187$$

$$H(z) = \frac{1 - (0.9048)(0.9899)z^{-1}}{1 - 2(0.9048)(-0.9899)z^{-1} + 0.8187z^{-2}}$$

$$H(z) = \frac{1 + 0.8956z^{-1}}{1 + 1.7913z^{-1} + 0.8187z^{-2}}$$

Method 2

The order of function is N

if $N = \text{odd}$.

$$s_n = e^{j\frac{\pi k}{N}}$$

$$k = 1 \text{ to } 2N.$$

if $N = \text{even}$.

$$s_n = e^{j\frac{(2k-1)\pi}{2N}}$$

Normalized Transfer function of the order when specification is given.

$$N=1 \rightarrow H(s_n) = \frac{1}{s_n + 1}$$

$$N=2 \rightarrow H(s_n) = \frac{1}{s_n^2 + 1.414s_n + 1}$$

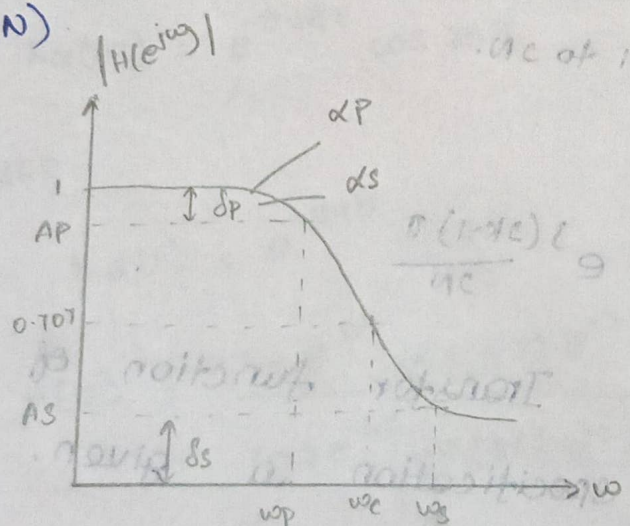
$$N=3 \rightarrow H(s_n) = \frac{1}{(s_n+1)(s_n^2+s_n+1)}$$

$$N=4 \rightarrow H(s_n) = \frac{1}{(s_n^2+0.765s_n+1)(s_n^2+1.848s_n+1)}$$

$$N=5 \rightarrow H(s_n) = \frac{1}{(s_n+1)(s_n^2+0.6185s_n+1)(s_n^2+1.618s_n+1)}$$

steps involved to design of digital IIR filter when its specifications are given.

- 1). N- specify the filter.
- 2). Determine the order of the filter (N).



where, $\omega_p \rightarrow$ digital passband frequency in rad/sec.

$A_p \rightarrow$ Passband gain at $\omega = \omega_p$.

$\omega_s \rightarrow$ digital stopband frequency in rad/sec.

$A_s \rightarrow$ s.B gain at $\omega = \omega_s$

$\alpha_p \rightarrow$ pass band attenuation.

$\delta_p \rightarrow$ pass band ripple.

$\alpha_s \rightarrow$ stop band attenuation

$\delta_s \rightarrow$ s.B ripple.

$\omega_c \rightarrow$ cutoff frequency (radian/sec) or

3dB frequency, Break over frequency, corner frequency
 $20 \log(10.707) = 3 \text{ dB}$.

cutoff frequency is the frequency at

which the transition of P.B to s.B occurs.

Relation between specifications:

i. $\alpha_p = \frac{1}{A_p}$

$$\alpha_p \text{ dB} = 20 \log_{10} \left(\frac{1}{A_p} \right)$$

$$= -20 \log_{10} (A_p) = -A_p \text{ dB}$$

$$A_p = 10^{(-\alpha_p \text{ dB} / 20)}$$

$$\alpha_p \text{ dB} = -A_p \text{ dB}$$

$$\Rightarrow A_p \text{ dB} = -\alpha_p \text{ dB}$$

ii. for stop band.

$$\alpha_s \text{ dB} = -A_s \text{ dB} \Rightarrow A_s \text{ dB} = -\alpha_s \text{ dB}$$

Since attenuation factor is same as ripple factor.

$$A_s \text{ (dB)} = -\delta_s \text{ dB}$$

$$A_p \text{ (dB)} = -\delta_p \text{ dB}$$

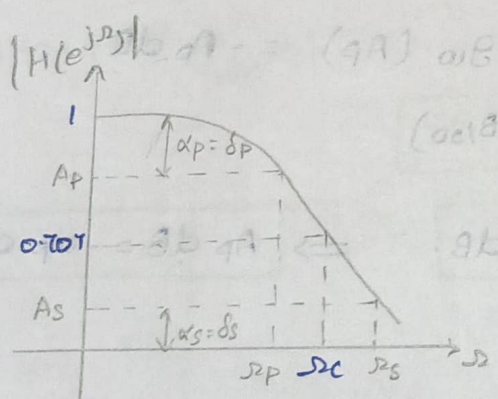
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Butterworth digital IIR filter.

characteristics:

1. All pole filter.
2. 'N' $\rightarrow$ order
3. Flat 'PB' & Flat 'SB' (there is no ripple in PB & SB)
4. No ripple in passband and stopband.
5. If $N \rightarrow \infty$, Magnitude response is ideal response.
6. $\omega_c \rightarrow$ gain of butterworth filter is 10.7 % of its max. gain.

steps involved in the design of digital IIR Butterworth filter.



where, $\omega_p \rightarrow$ analog passband frequency.

$\omega_s \rightarrow$ analog SB frequency

$\alpha_p \rightarrow$ passband attenuation

$\alpha_s \rightarrow$ SB

$A_p \rightarrow$ PB gain

$A_s \rightarrow$ SB

step-1: determine 'N'

$$N \geq \frac{\frac{1}{2} \log_{10} \left[\frac{\left(\frac{1}{A_s^2} \right)^{-1}}{\left(\frac{1}{A_p^2} \right)^{-1}} \right]}{\log_{10} \left[\frac{\omega_s}{\omega_p} \right]} \quad \text{or} \quad N \geq \frac{\frac{1}{2} \log_{10} \left[\frac{(0.1 \alpha_s) \text{ dB}}{(0.1 \alpha_p) \text{ dB}} \right]}{\log_{10} \left(\frac{\omega_s}{\omega_p} \right)}$$

$\therefore$ order must be always a +ve integer.

always round it to next highest integer

if $N = 0.1 \rightarrow 1$

$0.1 \rightarrow 3$

$0.4 \rightarrow 1$

$0.01 \rightarrow 31$ (or)

$0.9 \rightarrow 1$

step-2: determine the cutoff frequency ω_c

$$\omega_c = \frac{\omega_s}{\left(\left[\frac{1}{A_s^2} \right] - 1 \right)^{1/2N}} = \frac{\omega_s}{\left(10^{(0.1 \alpha_s) \text{ dB} / -1} - 1 \right)^{1/2N}}$$

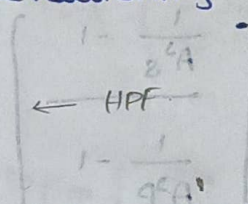
(or)

$$\omega_c = \frac{\omega_p}{\left(\left(\frac{1}{A_p^2} \right) - 1 \right)^{1/2N}} = \frac{\omega_p}{\left(10^{0.1 \alpha_p \text{ dB} / -1} - 1 \right)^{1/2N}}$$

step-3: choose the design desired normalized transfer function $H(s_n)$ based on the value of 'N'

step-4: Determine the actual transfer function $H(s)$ by evaluating $H(s_n)$ at $s_n = \frac{s}{\omega_c}$

at $s_n = \frac{s}{\omega_c}$



step-5: Determine $H(z)$ from $H(s)$ using

11V 1 BLT $\left(\frac{z-1}{z+1} \right)$

step-6: Find $H(e^{j\omega})$ by evaluating $H(z)$ at $z = e^{j\omega}$

step-7: Find $|H(e^{j\omega})| \rightarrow$ magnitude response.

Note:

If the specification is in digital terms
(ω) then we use this

$$\Delta T \rightarrow \omega = \Delta T$$

$$B\&T \rightarrow \Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \quad \text{or} \quad \omega = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right)$$

Find its analog frequencies (Ω)

- ① Design a Butterworth IIR digital LPF using Bilinear transformation technique by taking $T = 0.1 \text{ sec}$ to satisfy the following specification.

$$0.6 \leq |H(e^{j\omega})| \leq 1.0 \quad ; \text{ for } 0 \leq \omega \leq 0.35\pi$$

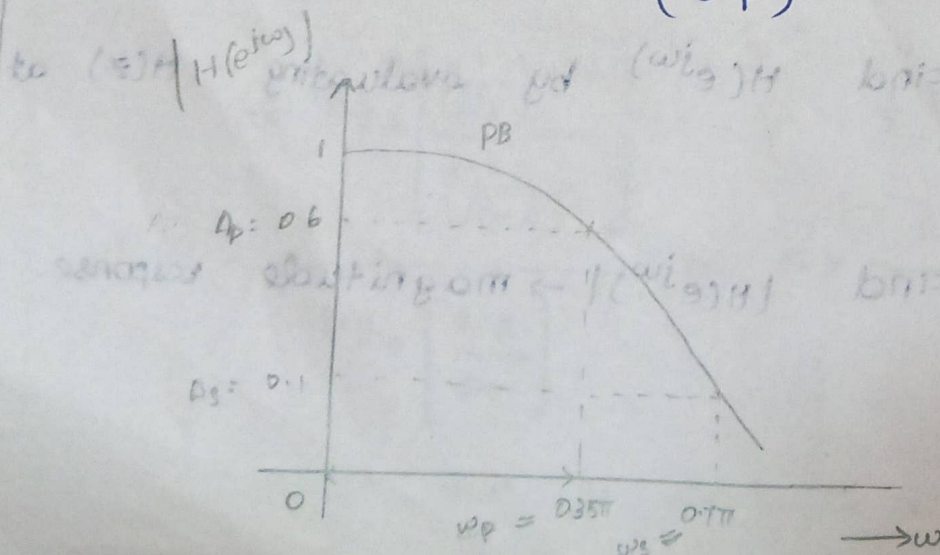
$$|H(e^{j\omega})| \leq 0.1 \quad ; \text{ for } 0.7\pi \leq \omega \leq \pi$$

Sol:

- ② to find order,

$$N \geq \frac{1}{2} \log_{10} \left[\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_p^2} - 1} \right]$$

$$\log_{10} \left(\frac{\Omega_s}{\Omega_p} \right)$$



①. H.K.T, $\Delta = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$

$\therefore T = 0.1, \omega_p = 0.35\pi$

$$\Delta_p = \frac{2}{0.1} \tan\left(\frac{0.35\pi}{2}\right)$$

$$= 20 \times 0.6128$$

$$\Delta_p = 12.256 \text{ rad/sec}$$

$$\Delta_s = \frac{2}{0.1} \tan\left(\frac{0.7\pi}{2}\right)$$

$$= 20 \times 1.9623$$

$$\Delta_s = 39.246 \text{ rad/sec}$$

$$N \geq \frac{1}{2} \log_{10} \left[\frac{\frac{1}{(0.6)^2_{0.01}} - 1}{\frac{1}{(0.6)^2_{0.36}} - 1} \right]$$

$$\log_{10} \left(\frac{39.246}{12.256} \right)$$

$$N \geq \frac{1}{2} \log_{10} \left(\frac{99}{1.777} \right)$$

$$\log_{10} (2.202)$$

$$N \geq \frac{1}{2} \times \frac{1.745}{0.605}$$

$$N \geq \frac{1.745}{1.01}$$

$$N \geq 1.727$$

$$\therefore N = 2$$

③ To find cutoff frequency.

$$\Omega_c = \frac{\Omega_p}{\left[\left(\frac{1}{A_p^2} \right) - 1 \right]^{\frac{1}{2N}}}$$

$$\therefore N=2, A_p^2=0.36, \Omega_p=12.556$$

$$\Omega_c = \frac{12.556}{(1.777)^{\frac{1}{4}}}$$

$$= \frac{12.556}{1.154}$$

$$\Omega_c = 10.876 \text{ rad/sec}$$

$$\Omega_c = \frac{\Omega_s}{\left[\left(\frac{1}{A_s^2} \right) - 1 \right]^{\frac{1}{2N}}}$$

$$= \frac{39.246}{(99)^{\frac{1}{4}}}$$

$$= \frac{39.246}{3.154}$$

$$\Omega_c = 12.443 \text{ rad/sec}$$

④ choose the normalized T.F based of order N

here, $N=2$.

$$H(s_n) = \frac{1}{s_n^2 + 1.414s_n + 1}$$

5. determine $H(s)$ by $H(s_n)$ for LPF

$$s_n = \frac{s}{\omega_c}$$

$$H(s) = \frac{1}{\left(\frac{s}{15.443}\right)^2 + 1.414 \left(\frac{s}{15.443}\right) + 1}$$

$$= \frac{1}{\frac{s^2}{154.828} + 0.11268 s + 1}$$

$$= \frac{154.828}{s^2 + (1.414 \times 15.443) s + (15.443)^2}$$

$$H(s) = \frac{154.828}{s^2 + 17.5944 s + 154.828}$$

$$H(z) = \frac{154.828}{s^2 + 17.5944 s + 154.828}$$

6. Determine $H(z)$ from $H(s)$ from BLT

$$H(z) = H(s) \quad \text{at} \quad s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$H(z) = \frac{154.828}{\left[\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + \frac{35.1888}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 154.828}$$

$$\text{at } T = 0.1$$

$$= \frac{154.828 (1+z^{-1})^2}{400 (1-z^{-1})^2 + 351.888 (1-z^{-1})(1+z^{-1}) + 154.828 (1+z^{-1})^2}$$

$$= \frac{154.828 (1) + 154.828 z^{-2} + 309.6564 z^{-1}}{400 + 400 z^{-2} - 800 z^{-1} + 351.888 - 351.888 z^{-2} + 154.828 + 154.828 z^{-2} + 309.6564 z^{-1}}$$

$$400 + 400 z^{-2} - 800 z^{-1} + 351.888 - 351.888 z^{-2} +$$

$$154.828 + 154.828 z^{-2} + 309.6564 z^{-1}$$

$$= \frac{154.8282 + 309.6564 z^{-1} + 154.8282 z^{-2}}{906.7762 + (-490.3436) z^{-1} + 202.940 z^{-2}}$$

$$= \frac{906.7762}{906.7762} \left[\frac{0.1707 + 0.3414 z^{-1} + 0.1707 z^{-2}}{1 - 0.5407 z^{-1} + 0.2238 z^{-2}} \right]$$

$$H(z) = \frac{0.1707 + 0.3414 z^{-1} + 0.1707 z^{-2}}{1 - 0.5407 z^{-1} + 0.2238 z^{-2}}$$

⊗ Find $H(e^{j\omega})$ by evaluating $H(z)$

at $e^{j\omega} = z$

$$H(e^{j\omega}) = \frac{0.1707 + 0.3414 e^{-j\omega} + 0.1707 e^{-2j\omega}}{1 - 0.5407 e^{-j\omega} + 0.2238 e^{-2j\omega}}$$

$$e^{-j\omega} = -0.5877 - j0.8090, \quad e^{-2j\omega} = -0.3090 + j0.9510$$

$$0.1707 + 0.3414 (-0.5877 - j0.8090) + 0.1707$$

$$\left(\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right) \frac{333.188}{T} + \left(\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right) \frac{C}{T} (-0.3090 + j0.9510)$$

$$1 - 0.5407 (-0.5877 - j0.8090) + 0.2238$$

$$(-0.3090 + j0.9510)$$

$$0.1707 - 0.2006 - j0.2761 + 0.0527 + j0.1623$$

$$1 + 0.3177 + j0.4374 - 0.0691 + j0.2128$$

$$H(e^{j\omega}) = \frac{-0.0826 - j0.1138}{1.2486 + j0.6502}$$

By taking Magnitude.

$$|H(e^{j\omega})| = \sqrt{\frac{(-0.0826)^2 + (-0.1138)^2}{(1.2486)^2 + (0.6502)^2}}$$

$$= \sqrt{\frac{6.8227 \times 10^{-3} + 0.0129}{1.5590 + 0.4227}}$$

$$= \sqrt{\frac{0.0197}{1.9817}}$$

$$= \sqrt{9.940 \times 10^{-3}}$$

$$|H(e^{j\omega})| = 0.099$$

$$\boxed{|H(e^{j\omega})| \approx 0.1}$$

②. Design a Butterworth digital IIR LPF using Bilinear Transformation (BLT) by taking $T = 0.1$ s to satisfy the following specification.

$$\delta_p = 4.43 \text{ dB}$$

$$\delta_s = 20 \text{ dB}$$

$$\omega_p = 0.35 \pi$$

$$\omega_s = 0.7 \pi$$

Sol:

$$A_p = 10^{\frac{-4.43}{20}}$$

$$= 10^{-0.2215}$$

$$A_p = 0.600$$

$$A_s = 10^{\frac{-20}{20} \text{ dB}}$$

$$= 10^{-1}$$

$$= 10^{-1}$$

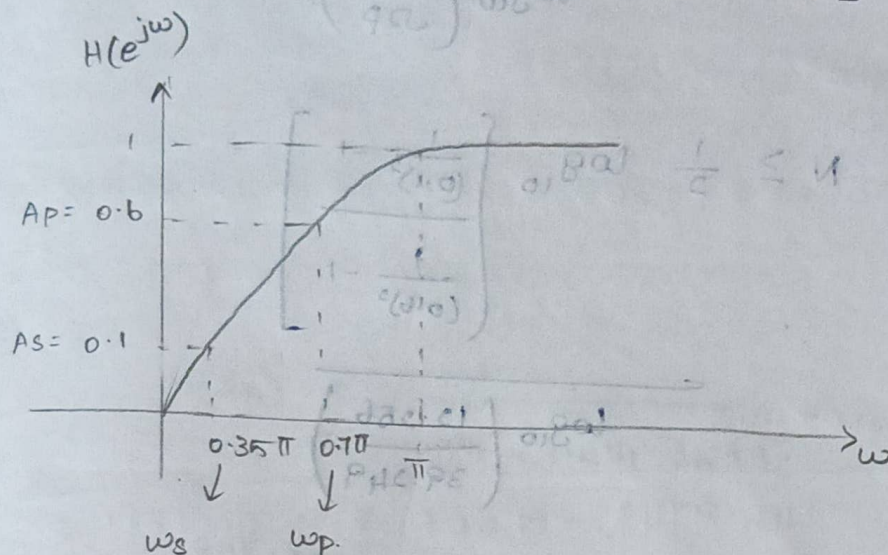
$$A_s = 0.1$$

refer question ①.

③ Design butterworth digital IIR HPF using BLT by taking $T = 0.18$ to satisfy the following specification.

$$0.6 \leq |H(e^{j\omega})| \leq 1 ; \text{ for } 0.7\pi \leq \omega \leq \pi$$

$$|H(e^{j\omega})| \leq 0.1 ; \text{ for } 0 \leq \omega \leq 0.35\pi.$$



①. To find equivalent analog frequency.

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

$$\therefore \omega_p = 0.7\pi$$

$$= \frac{2}{0.18} \tan\left(\frac{0.7\pi}{2}\right)$$

$$= 20 \times 1.9624$$

$$\Omega_p = 39.249 \text{ rad/sec.}$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right)$$

$$= \frac{2}{0.18} \tan\left(\frac{0.35\pi}{2}\right)$$

$$= 20 \times 0.6128$$

$$\Omega_s = 12.2560 \text{ rad/sec}$$

②. Find the order.

$$N \geq \frac{1}{2} \log_{10} \left(\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_p^2} - 1} \right)$$

$$\log_{10} \left(\frac{\Omega_s}{\Omega_p} \right)$$

$$N \geq \frac{1}{2} \log_{10} \left[\frac{\frac{1}{(0.1)^2} - 1}{\frac{1}{(0.6)^2} - 1} \right]$$

$$\log_{10} \left(\frac{13.256}{39.249} \right)$$

$$N \geq \frac{1}{2} \log_{10} \left(\frac{99}{1.777} \right) \xrightarrow{55.7118}$$

$$\log_{10} (0.3122)$$

$$N \geq \frac{1}{2} \frac{1.7459}{-0.505}$$

$$N \geq \frac{-1.7459}{1.011}$$

$$N \geq -1.726$$

$$\boxed{N = 2}$$

③. Find cutoff frequency.

$$\Omega_c = \frac{\Omega_s}{\left(\frac{1}{A_s^2} - 1 \right)^{1/2N}}$$

$$\Omega_c = \frac{15.256}{\left(\frac{1}{(0.1)^2} - 1\right)^{1/4}}$$

$$= \frac{15.256}{(6992)^{1/4}}$$

$$= \frac{15.256}{3.1543}$$

$$\Omega_c = 3.8854 \text{ rad/sec.}$$

$$3.8854$$

(11)

$$H(s_n) = \frac{1}{s_n^2 + 1.414 s_n + 1}$$

(12)

$$s_n = \frac{\Omega_c}{s} = \frac{3.8854}{s}$$

$$H(s) = \frac{1}{\left(\frac{3.8854}{s}\right)^2 + 1.414 \left(\frac{3.8854}{s}\right) + 1}$$

$$H(s) = \frac{s^2}{s^2 + 5.4939s + 15.0963}$$

(13)

$$H(z) = H(s) \quad \text{at} \quad s = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H(z) = \frac{\left(\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right)^2}{\left(\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right)^2 + 10.9878 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 15.0963}$$

$$\left(\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right)^2 + 10.9878 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 15.0963$$

$$T = 0.18$$

$$H(z) = \frac{400 \cdot \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 (1+z^{-1})^2}{400 (1-z^{-1})^2 + 109.878 (1-z^{-1})(1+z^{-1}) + 15.0963 (1+z^{-1})^2}$$

$$= \frac{400 + 400 z^{-2} - 800 z^{-1}}{400 + 400 z^{-2} - 800 z^{-1} + 109.878 - 109.878 z^{-2} + 15.0963 + 15.0963 z^{-2} + 30.1926 z^{-1}}$$

$$H(z) = \frac{400 - 800 z^{-1} + 400 z^{-2}}{524.9743 - 769.8074 z^{-1} + 305.2183 z^{-2}}$$

$$H(z) = \frac{0.7619 - 1.5238 z^{-1} + 0.7619 z^{-2}}{1 - 1.4663 z^{-1} + 0.5813 z^{-2}}$$

④. $z = e^{-j\omega}$

$$H(e^{-j\omega}) = \frac{0.7619 - 1.5238 (e^{-j\omega}) + 0.7619 (e^{-j2\omega})}{1 - 1.4663 (e^{-j\omega}) + 0.5813 (e^{-j2\omega})}$$

$$e^{-j\omega_3} = 0.4539 - j0.8910, \quad e^{-j2\omega_3} = -0.5877 - j0.8090$$

$$= \frac{0.7619 - 1.5238 (0.4539 - j0.8910) + 0.7619 (-0.5877 - j0.8090)}{1 - 1.4663 (0.4539 - j0.8910) + 0.5813 (-0.5877 - j0.8090)}$$

$$1 - 1.4663 (0.4539 - j0.8910) + 0.5813 (-0.5877 - j0.8090)$$

$$H(e^{j\omega}) = \frac{0.7619 - 0.6916 + j1.3577 - 0.4477 - j0.6163}{1 - 0.6655 + j1.3064 - 0.3046 - j0.4193}$$

$$H(e^{j\omega}) = \frac{-0.8774 + j0.7414}{0.0299 + j0.8871}$$

⑧. Magnitude.

$$|H(e^{j\omega})| = \sqrt{\frac{(-0.8774)^2 + (0.7414)^2}{(0.0299)^2 + (0.8871)^2}}$$

$$= \sqrt{\frac{0.1424 + 0.5496}{(8.9401 \times 10^{-4}) + 0.7869}}$$

$$= \sqrt{\frac{0.692}{0.7877}}$$

$$= \sqrt{0.8785}$$

$$|H(e^{j\omega})| = 0.9372$$

Design a butterworth digital FIR LPF using BLT by taking $T=1$ s to satisfy the following specification.

$$PB \text{ ripple} \leq 3.011 \text{ dB}$$

$$SB \text{ attenuation} \geq 21.9382 \text{ dB.}$$

$$PB \text{ edge frequency} = 0.2\pi \text{ rad/sample.}$$

$$SB \text{ edge frequency} = 0.4\pi \text{ rad/sample.}$$

Sol: $\delta_p = 8.011 \text{ dB}$

$\alpha_s = 21.9382 \text{ dB}$

$\omega_p = 0.2\pi \text{ rad/sample}$

$\omega_s = 0.4\pi \text{ " "}$

$A_s = 10^{-(\alpha_s \text{ dB}/20)}$

$= 10^{-(21.9382/20)}$

$= 10^{-1.0969}$

$A_s = 0.080$

$= 10^{-(3.011/20)}$

$A_p = 10$

$A_p = 0.707$

$\alpha_p \text{ dB} = -\delta_p \text{ dB}$

$\alpha_p = -3.011$

1. To find analog frequency,

$\omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$

$= \frac{2}{1} \tan\left(\frac{0.2\pi}{2}\right)$

$= 2 \times 0.3249$

$\omega_p = 0.6498 \text{ rad/sam.}$

$\omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right)$

$= \frac{2}{1} \tan\left(\frac{0.4\pi}{2}\right)$

$\omega_s = 1.4530 \text{ rad/sam.}$

2). order.

$$N \geq \frac{1}{2} \log_{10} \left[\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_P^2} - 1} \right]$$

$$\log_{10} \left[\frac{\Sigma_S}{\Sigma_P} \right]$$

$$N \geq \frac{1}{2} \log_{10} \left[\frac{\frac{1}{(0.08)^2} - 1}{\frac{1}{(0.101)^2} - 1} \right]$$

$$\log_{10} \left[\frac{1.4530}{0.6498} \right]$$

$$N \geq \frac{1}{2} \log_{10} \left[\frac{156.25 - 1}{2.0008 - 1} \right]$$

$$\log_{10} (2.2860)$$

$$N \geq \frac{1}{2} \log \left(\frac{156.25}{1.0008} \right)$$

$$0.3494$$

$$N \geq \frac{2.1906}{0.6988}$$

$$N \geq 3.1248$$

$$\boxed{N = 4}$$

3). To find cut-off frequency,

$$\Omega_c = \frac{\Omega_s}{\left(\frac{1}{A_s^2} - 1 \right)^{1/2N}} \quad \therefore N=4,$$

$$= \frac{1.4530}{\left(\frac{1}{(0.08)^2} - 1 \right)^{1/8}}$$

$$= \frac{1.4530}{(155.25)^{1/8}} = \frac{1.4530}{1.8787}$$

$$\boxed{\Omega_c = 0.7734 \text{ rad/sec.}}$$

4). $N=4,$

$$H(s_n) = \frac{1}{(s_n^2 + 0.765s_n + 1)(s_n^2 + 1.848s_n + 1)}$$

5). $s_n = \frac{s}{\Omega_c}$ (LPE)

$$H(s) = \frac{1}{\left[\left(\frac{s}{0.7734} \right)^2 + 0.765 \left(\frac{s}{0.7734} \right) + 1 \right] \left[\left(\frac{s}{0.7734} \right)^2 + \frac{1.848s}{0.7734} + 1 \right]}$$

$$= \frac{0.5981 \times 0.5981}{[s^2 + (0.765 \times 0.7734)s + 1] [s^2 + (1.848 \times 0.7734)s + 1]}$$

$$= \frac{0.3577}{(s^2 + 0.5916s + 1)(s^2 + 1.4292s + 1)}$$

6).

$$H(z) = HCS)$$

$$\text{at } S = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H(z) = \frac{0.3577}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.5916 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.5981 \right] (1-z^{-2})}$$

$$\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 1.4292 \times 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.5981 \right]$$

$$= \frac{0.3577 (1+z^{-1})^2 (1+z^{-1})^2}{(4+4z^{-2}-8z^{-1} + 1.1832 - 1.1832z^{-2} + 0.5981 + 0.5981z^{-2} + 1.1962z^{-1})}$$

$$(4+4z^2-8z^{-1} + 2.8584 - 2.8584z^{-2} + 0.5981 + 0.5981z^{-2} + 1.1962z^{-1})$$

$$= \frac{0.3577 (1+z^{-2}+2z^{-1}) (1+2z^{-1}+z^{-2})}{(15.7813 - 6.8038z^{-1} + 3.4149z^{-2})}$$

$$(7.4565 - 6.8038z^{-1} + 1.7397z^{-2})$$

$$= 0.3577 (1+2z^{-1}+z^{-2} + z^{-2}+2z^{-3}+z^{-4} + 2z^{-1}+4z^{-2}+2z^{-3})$$